

What is Networking?

Learning Objectives

- Define computer networking and its purpose
- Identify real-world examples of networks
- Explain how devices communicate through a network

Timing Breakdown

0-5 min	Engage	Ask: 'How do you search for something on Google? What makes one search better th...
5-15 min	Explore	Give each pair a search challenge: 'Find the population of Chitradurga district....
15-25 min	Explain	Define email components: address format username\@domain, providers Gmail, Yah...
25-35 min	Elaborate	Task: 'Compose an email to your teacher requesting permission for a school trip....
35-40 min	Evaluate	Exit ticket: '1. Name two search tips. 2. What is the format of an email address...

Engage (5-7 minutes)

Teacher Activity

Ask: 'How do you search for something on Google? What makes one search better than another?' Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: 'Find the population of Chitradurga district.' Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Search results use ranking algorithms based on relevance scores

Science: Internet infrastructure uses physical cables and wireless signals — physics of communication

Language: Formal email writing follows professional communication conventions

Digital Citizen Reminder: Think before you click Send — emails are permanent. Be respectful and professional.

Dormitory Task (Paper-and-Pencil)

Write a formal email on paper to your principal requesting a computer lab upgrade. Include subject, greeting, 3 reasons, and proper closing. Then write a search query you would use to find 'best computers for schools under Rs 50,000'.

Resources Needed:

- Computer with internet
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Differentiation

Approaching: Use a search tutorial guide and email template with blanks

On Level: Search independently and compose email as described

Advanced: Compare search results from 3 engines and write an evaluation report

SEND: Guided search with teacher supervision, email template with sentence starters

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Evaluation — judging the quality and reliability of search results

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Types of Networks

Learning Objectives

- Differentiate between LAN, MAN, and WAN
- Give examples of each network type from daily life
- Understand the range and speed differences between network types

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Network Devices

Learning Objectives

- Identify common network devices and their functions
- Explain the role of routers, switches, and modems
- Differentiate between wired and wireless devices

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Connection to next lesson:

Internet Connection

Learning Objectives

- Identify different types of internet connections
- Compare broadband, mobile data, and Wi-Fi
- Understand the concept of bandwidth and speed

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Teacher Reflection (Post-Lesson)

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Connection to next lesson:

Internet Protocols

Learning Objectives

- Define what an internet protocol is
- Explain the role of IP addresses and URLs
- Understand how HTTP and HTTPS work

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Email Services

Learning Objectives

- Understand what email is and how it works
- Identify the parts of an email address
- Compose a basic email with subject, body, and recipient

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Attachments

Learning Objectives

- Attach files to an email correctly
- Understand file size limits and common formats
- Open and save email attachments safely

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Teacher Reflection (Post-Lesson)

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Connection to next lesson:

Internet Safety

Learning Objectives

- Identify common online threats and scams
- Practice safe browsing habits
- Understand the importance of strong passwords

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Social Media

Learning Objectives

- Identify popular social media platforms
- Understand the benefits and risks of social media
- Practice responsible social media behavior

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Connection to next lesson:

Download Management

Learning Objectives

- Download files safely from the internet
- Understand file types and where downloads are saved
- Organize and manage downloaded files

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Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: 'Find the population of Chitradurga district.' Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Search results use ranking algorithms based on relevance scores

Science: Internet infrastructure uses physical cables and wireless signals — physics of communication

Language: Formal email writing follows professional communication conventions

Digital Citizen Reminder: Think before you click Send — emails are permanent. Be respectful and professional.

Dormitory Task (Paper-and-Pencil)

Write a formal email on paper to your principal requesting a computer lab upgrade. Include subject, greeting, 3 reasons, and proper closing. Then write a search query you would use to find 'best computers for schools under Rs 50,000'.

Resources Needed:

- Computer with internet
- email access

- search challenge cards
- email template

Differentiation

Approaching: Use a search tutorial guide and email template with blanks

On Level: Search independently and compose email as described

Advanced: Compare search results from 3 engines and write an evaluation report

SEND: Guided search with teacher supervision, email template with sentence starters

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Evaluation — judging the quality and reliability of search results

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Network Troubleshooting

Learning Objectives

- Identify common network problems
- Use basic troubleshooting steps
- Understand when to seek technical help

Timing Breakdown

0-5 min	Engage	Ask: 'How do you search for something on Google? What makes one search better th...
5-15 min	Explore	Give each pair a search challenge: 'Find the population of Chitradurga district....
15-25 min	Explain	Define email components: address format username\@domain, providers Gmail, Yah...
25-35 min	Elaborate	Task: 'Compose an email to your teacher requesting permission for a school trip....
35-40 min	Evaluate	Exit ticket: '1. Name two search tips. 2. What is the format of an email address...

Engage (5-7 minutes)

Teacher Activity

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Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 1 Review

Learning Objectives

- Review all Chapter 1 networking concepts
- Connect networking topics to real-world applications
- Identify areas needing further practice

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to GIMP

Learning Objectives

- Open and navigate the GIMP interface
- Identify key toolbox elements and menus
- Create and save a new image file

Timing Breakdown

0-5 min	Engage	Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What ch...
5-15 min	Explore	Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students...
15-25 min	Explain	Define layers: transparent sheets stacked to create the final image. Each layer ...
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35-40 min	Evaluate	Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why a...

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

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Answer the questions. Show their work to a partner.

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Dormitory Task (Paper-and-Pencil)

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SEND: Pre-made layer templates to modify, guided steps with visual aids

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Algorithm Design — planning editing steps before executing in the software

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Selection Tools

Learning Objectives

- Use rectangle, ellipse, and free select tools
- Understand when to use each selection type
- Practice selecting and modifying portions of an image

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Layers Concept

Learning Objectives

- Understand what layers are and why they matter
- Create, rename, and reorder layers
- Use layer visibility and opacity controls

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Layer Effects

Learning Objectives

- Apply drop shadows and glows to layers
- Use layer masks for non-destructive editing
- Combine multiple layer effects for creative results

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Drawing Tools

Learning Objectives

- Use pencil, paintbrush, and airbrush tools
- Adjust brush size, shape, and opacity
- Create freehand drawings and simple illustrations

Timing Breakdown

0-5 min	Engage	Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What ch...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Color Adjustments

Learning Objectives

- Adjust brightness, contrast, and color balance
- Use hue-saturation to modify colors
- Correct common photo problems like underexposure

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Filters Introduction

Learning Objectives

- Understand what filters do to images
- Apply blur, sharpen, and noise filters
- Use the preview feature before applying filters

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Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Pixel dimensions use multiplication; aspect ratios involve proportional reasoning

Science: Color mixing in digital images follows additive light physics RGB

Language: Visual design principles enhance communication clarity

Digital Citizen Reminder: Only use images you have permission to edit — respect copyright and others' work.

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Differentiation

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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Algorithm Design — planning editing steps before executing in the software

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Artistic Filters

Learning Objectives

- Apply oil painting and watercolor effects
- Use the cartoon filter for stylized images
- Combine multiple artistic filters creatively

Timing Breakdown

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5-15 min	Explore	Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students...
15-25 min	Explain	Define layers: transparent sheets stacked to create the final image. Each layer ...
25-35 min	Elaborate	Task: 'Create a simple composition with background on one layer and text on anot...
35-40 min	Evaluate	Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why a...

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Text and Typography

Learning Objectives

- Add and format text on images
- Change font, size, color, and alignment
- Apply text effects like shadows and outlines

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Image Composition

Learning Objectives

- Combine multiple images into one composition
- Use cropping and resizing for proper layout
- Apply the rule of thirds for balanced design

Timing Breakdown

0-5 min	Engage	Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What ch...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Photo Editing

Learning Objectives

- Apply all learned GIMP skills to edit a photo
- Practice selection, layers, filters, and text together
- Present a completed photo editing project

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 2 Review

Learning Objectives

- Review all Chapter 2 GIMP concepts
- Connect graphics skills to real-world applications
- Identify areas needing further practice

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

What is Multimedia?

Learning Objectives

- Define multimedia and its components
- Identify text, audio, image, video, and animation as media types
- Explain how multimedia is used in daily life

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
15-25 min	Explain	Define video communication: sharing information through moving images and sound....
25-35 min	Elaborate	Task: 'Plan a 1-minute video about your school. Write the script, list shots nee...
35-40 min	Evaluate	Exit ticket: '1. What are the steps to create a video? 2. Name two video file fo...

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea → script → shoot → edit → export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Video resolution uses pixel dimensions; frame rate involves counting per second

Science: Video compression uses mathematical algorithms to reduce file size

Language: Storytelling through video requires clear narrative structure

Digital Citizen Reminder: Do not use copyrighted videos without permission — use free resources or create your own.

Dormitory Task (Paper-and-Pencil)

Create a storyboard for a 30-second video about your village. Draw 6 frames like a comic strip, write narration for each, and plan where transitions would go.

Resources Needed:

- Computer with OpenShot if available
- storyboard template
- sample video clips
- storyboard paper

Differentiation

Approaching: Use a pre-made storyboard template with guidance

On Level: Create a video plan with script and storyboard independently

Advanced: Edit a short video with transitions, titles, and background music

SEND: Draw storyboard frames with picture prompts, guided narration writing

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Abstraction — selecting key moments from a longer story for a short video

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Audio Editing with Audacity

Learning Objectives

- Open Audacity and identify its interface components
- Record, play, and save a short audio clip
- Apply basic trimming and cutting operations on audio

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
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Method: Exit ticket and self-assessment

CT Skill Assessed: Abstraction — selecting key moments from a longer story for a short video

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Audio Mixing

Learning Objectives

- Import multiple audio tracks into Audacity
- Adjust volume levels and pan left/right channels
- Mix multiple tracks and export the final audio

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Video Concepts

Learning Objectives

- Define video and explain how frames create motion
- Identify video formats and their uses
- Describe the difference between analog and digital video

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Video Editing Basics

Learning Objectives

- Import video clips into a video editor
- Trim, split, and arrange clips on a timeline
- Add text overlays and export the edited video

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
15-25 min	Explain	Define video communication: sharing information through moving images and sound....
25-35 min	Elaborate	Task: 'Plan a 1-minute video about your school. Write the script, list shots nee...
35-40 min	Evaluate	Exit ticket: '1. What are the steps to create a video? 2. Name two video file fo...

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea → script → shoot → edit → export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Video resolution uses pixel dimensions; frame rate involves counting per second

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Language: Storytelling through video requires clear narrative structure

Digital Citizen Reminder: Do not use copyrighted videos without permission — use free resources or create your own.

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Differentiation

Approaching: Use a pre-made storyboard template with guidance

On Level: Create a video plan with script and storyboard independently

Advanced: Edit a short video with transitions, titles, and background music

SEND: Draw storyboard frames with picture prompts, guided narration writing

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Abstraction — selecting key moments from a longer story for a short video

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Presentation Software

Learning Objectives

- Identify features of presentation software like LibreOffice Impress
- Create a new presentation with multiple slides
- Add text, images, and shapes to slides

Timing Breakdown

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5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
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Formative Assessment

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Slide Design

Learning Objectives

- Apply consistent themes and backgrounds to slides
- Use fonts, colors, and alignment for readability
- Create visually appealing slide layouts

Timing Breakdown

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5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Animations

Learning Objectives

- Apply entrance, emphasis, and exit animations to objects
- Set animation order and timing
- Use animations effectively without overdoing them

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Slide Transitions

Learning Objectives

- Apply different transition effects between slides
- Set transition duration and advance timing
- Choose appropriate transitions for different contexts

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Formative Assessment

Method: Exit ticket and self-assessment

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Multimedia Integration

Learning Objectives

- Embed audio and video clips into presentations
- Insert hyperlinks and interactive buttons
- Create a cohesive multimedia presentation

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Abstraction — selecting key moments from a longer story for a short video

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Presentation

Learning Objectives

- Plan and design a complete multimedia presentation
- Apply all learned skills: design, animation, transitions, and multimedia
- Present and evaluate peer presentations effectively

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
5-15 min	Explore	Show OpenShot interface or describe it. Demonstrate: import media, add to time...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 3 Review

Learning Objectives

- Recall all multimedia concepts from Chapter 3
- Demonstrate audio and video editing skills
- Apply presentation design and multimedia integration

Timing Breakdown

0-5 min	Engage	Show a short video clip 30 seconds. Ask: 'How do videos communicate messages b...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Spreadsheets

Learning Objectives

- Define spreadsheet and its purpose in data management
- Identify rows, columns, cells, and cell references
- Open LibreOffice Calc and navigate the interface

Timing Breakdown

0-5 min	Engage	Show a poorly formatted document vs a professionally formatted one. Ask: 'Which ...
5-15 min	Explore	Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets ...
15-25 min	Explain	Explain document structuring: headings create hierarchy, bullets organize lists,...
25-35 min	Elaborate	Task: 'Create a class notice about a school event. Include: heading, date, detai...
35-40 min	Evaluate	Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3....

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Table formatting uses row/column organization; alignment involves spatial reasoning

Science: Scientific papers follow strict formatting conventions for clarity

Language: Document structure mirrors essay organization: introduction, body, conclusion

Digital Citizen Reminder: Always proofread before printing — paper waste is expensive and unnecessary.

Dormitory Task (Paper-and-Pencil)

On paper, design a class notice about a sports day. Include a heading, date, 3 bullet points with details, and a schedule table with 4 rows. Use proper alignment and spacing.

Resources Needed:

- Computer with LibreOffice Writer
- sample documents
- notice template
- checklist

Differentiation

Approaching: Use a pre-formatted template and fill in content

On Level: Create the notice independently with all formatting elements

Advanced: Create a multi-page newsletter with images, columns, and table of contents

SEND: Simplified template with visual guides, step-by-step formatting instructions

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Decomposition — breaking document creation into layout, content, and formatting steps

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Basic Formulas

Learning Objectives

- Write formulas using addition, subtraction, multiplication, and division
- Use the equals sign to start formulas
- Understand the difference between formula and value

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Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Decomposition — breaking document creation into layout, content, and formatting steps

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Cell References

Learning Objectives

- Distinguish between relative, absolute, and mixed references

Timing Breakdown

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Engage (5-7 minutes)

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Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

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Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: ‘How do you send the same letter to 40 students with different names?’

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

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What worked well:

What to improve:

Connection to next lesson:

Functions

Learning Objectives

- Use SUM, AVERAGE, COUNT, MAX, and MIN functions
- Enter functions using the function wizard
- Apply functions to real-world data sets

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Charts Creation

Learning Objectives

- Select data ranges for chart creation
- Create bar, line, and pie charts
- Choose appropriate chart types for different data

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chart Formatting

Learning Objectives

- Modify chart titles, labels, and legends
- Change chart colors and styles
- Add trendlines and data labels to charts

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Sorting

Learning Objectives

- Sort data in ascending and descending order
- Sort by single and multiple columns
- Use sort to organize records meaningfully

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Filtering

Learning Objectives

- Apply auto-filter to display specific records
- Use filter criteria to show data meeting conditions
- Clear filters and understand filtered vs hidden data

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Validation

Learning Objectives

- Set validation rules for data entry in cells
- Create dropdown lists for restricted choices
- Display error messages for invalid entries

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Conditional Formatting

Learning Objectives

- Apply color scales to visualize data ranges
- Use data bars and icon sets for quick analysis
- Create custom rules for highlighting specific cells

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Grade Sheet

Learning Objectives

- Create a complete grade sheet using all spreadsheet skills
- Apply formulas, functions, charts, and formatting
- Analyze and present data findings effectively

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 4 Review

Learning Objectives

- Recall all data handling concepts from Chapter 4
- Demonstrate proficiency in formulas, functions, and charts
- Apply data analysis skills to solve real-world problems

Timing Breakdown

0-5 min	Engage	Show a poorly formatted document vs a professionally formatted one. Ask: 'Which ...
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Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Table formatting uses row/column organization; alignment involves spatial reasoning

Science: Scientific papers follow strict formatting conventions for clarity

Language: Document structure mirrors essay organization: introduction, body, conclusion

Digital Citizen Reminder: Always proofread before printing — paper waste is expensive and unnecessary.

Dormitory Task (Paper-and-Pencil)

On paper, design a class notice about a sports day. Include a heading, date, 3 bullet points with details, and a schedule table with 4 rows. Use proper alignment and spacing.

Resources Needed:

- Computer with LibreOffice Writer
- sample documents
- notice template
- checklist

Differentiation

Approaching: Use a pre-formatted template and fill in content

On Level: Create the notice independently with all formatting elements

Advanced: Create a multi-page newsletter with images, columns, and table of contents

SEND: Simplified template with visual guides, step-by-step formatting instructions

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Decomposition — breaking document creation into layout, content, and formatting steps

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Scratch Review

Learning Objectives

- Recall Scratch interface components and their functions
- Identify sprites, costumes, and backdrops in projects
- Navigate the Scratch programming environment confidently

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Game logic uses coordinates, angles, random numbers, and score calculations

Science: Physics of falling objects can be simulated with gravity variables

Language: Clear variable naming and code comments improve program readability

Digital Citizen Reminder: Share your Scratch projects responsibly — give credit to original creators.

Dormitory Task (Paper-and-Pencil)

On paper, write the pseudocode for a simple game: 'Guess the Number.' Include: start, generate random number, ask for guess, compare, give hint higher/lower, repeat until correct, end. Use a flowchart if preferred.

Resources Needed:

- Computer with Scratch
- Catch the Apples project
- timer worksheet
- pseudocode template

Differentiation

Approaching: Follow a step-by-step guide to add the timer feature

On Level: Add the timer feature independently as described

Advanced: Add multiple levels, difficulty settings, and a high score system

SEND: Paper-based block sequencing, pre-made scripts to arrange in order

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Variables Advanced

Learning Objectives

- Create and use variables in Scratch projects
- Understand the difference between local and global variables
- Implement variables for score tracking and game mechanics

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

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Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

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Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

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On Level: Add the timer feature independently as described

Advanced: Add multiple levels, difficulty settings, and a high score system

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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Conditional Logic

Learning Objectives

- Understand if-then and if-then-else blocks in Scratch
- Create programs that make decisions based on conditions
- Apply conditional logic to real-world problem scenarios

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
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25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

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Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

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Teacher Activity

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Elaborate (8-10 minutes)

Teacher Activity

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Student Activity

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Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

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Advanced: Add multiple levels, difficulty settings, and a high score system

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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Loops Advanced

Learning Objectives

- Use repeat, forever, and repeat-until loops in Scratch
- Understand when to use different loop types
- Create animated sequences using loops effectively

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
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Engage (5-7 minutes)

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Elaborate (8-10 minutes)

Teacher Activity

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Student Activity

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Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

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SEND: Paper-based block sequencing, pre-made scripts to arrange in order

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Sensing Blocks

Learning Objectives

- Explore Scratch sensing blocks for input detection
- Use distance, color, and key-press sensors in programs
- Create interactive projects that respond to user actions

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
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35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

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Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

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Teacher Activity

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Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

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Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

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On Level: Add the timer feature independently as described

Advanced: Add multiple levels, difficulty settings, and a high score system

SEND: Paper-based block sequencing, pre-made scripts to arrange in order

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Sprite Interactions

Learning Objectives

- Create multiple sprites that interact with each other
- Use broadcasts to coordinate sprite communication
- Design scenarios where sprites work together to complete tasks

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Game logic uses coordinates, angles, random numbers, and score calculations

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Language: Clear variable naming and code comments improve program readability

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Dormitory Task (Paper-and-Pencil)

On paper, write the pseudocode for a simple game: 'Guess the Number.' Include: start, generate random number, ask for guess, compare, give hint higher/lower, repeat until correct, end. Use a flowchart if preferred.

Resources Needed:

- Computer with Scratch
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Differentiation

Approaching: Follow a step-by-step guide to add the timer feature

On Level: Add the timer feature independently as described

Advanced: Add multiple levels, difficulty settings, and a high score system

SEND: Paper-based block sequencing, pre-made scripts to arrange in order

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Clones

Learning Objectives

- Understand what clones are and how they work in Scratch
- Create and delete clones dynamically during program execution
- Use clones to build multi-character game scenarios

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
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Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

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Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

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Answer the questions. Show their timer feature to a partner.

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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Game Design

Learning Objectives

- Understand core game mechanics: objectives, rules, and feedback
- Apply game design principles to create engaging gameplay
- Balance difficulty and reward in game elements

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
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Formative Assessment

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CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Keyboard Controls

Learning Objectives

- Implement keyboard input controls for sprite movement
- Create responsive controls using arrow keys and other inputs
- Design smooth movement systems for game characters

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
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Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Game logic uses coordinates, angles, random numbers, and score calculations

Science: Physics of falling objects can be simulated with gravity variables

Language: Clear variable naming and code comments improve program readability

Digital Citizen Reminder: Share your Scratch projects responsibly — give credit to original creators.

Dormitory Task (Paper-and-Pencil)

On paper, write the pseudocode for a simple game: 'Guess the Number.' Include: start, generate random number, ask for guess, compare, give hint higher/lower, repeat until correct, end. Use a flowchart if preferred.

Resources Needed:

- Computer with Scratch
- Catch the Apples project
- timer worksheet
- pseudocode template

Differentiation

Approaching: Follow a step-by-step guide to add the timer feature

On Level: Add the timer feature independently as described

Advanced: Add multiple levels, difficulty settings, and a high score system

SEND: Paper-based block sequencing, pre-made scripts to arrange in order

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Random Events

Learning Objectives

- Use random number generation in Scratch programs
- Create unpredictable game elements using randomness
- Understand how randomness adds variety and replayability

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

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Formative Assessment

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CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Game

Learning Objectives

- Apply all learned Scratch concepts to create a complete game
- Integrate variables, loops, conditionals, and sprite interactions
- Document and present a finished game project to peers

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
25-35 min	Elaborate	Task: 'Add a new feature to the game — a timer that counts down from 60 seconds....
35-40 min	Evaluate	Exit ticket: '1. What event starts the game? 2. How does the score variable incr...

Engage (5-7 minutes)

Teacher Activity

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Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

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Formative Assessment

Method: Exit ticket and self-assessment

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 5 Review

Learning Objectives

- Review and consolidate all Scratch concepts from Chapter 5
- Demonstrate mastery through a comprehensive review activity
- Identify areas for further practice and improvement

Timing Breakdown

0-5 min	Engage	Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow ...
5-15 min	Explore	Open the game project in Scratch. Show the scripts for each sprite. Let students...
15-25 min	Explain	Review Scratch concepts: events when key pressed, loops forever, repeat, con...
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Engage (5-7 minutes)

Teacher Activity

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Student Activity

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Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

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Elaborate (8-10 minutes)

Teacher Activity

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Student Activity

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Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

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Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Debugging — testing the timer feature and fixing any errors in the logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

What is Digital Citizenship?

Learning Objectives

- Define digital citizenship and its importance in modern life
- Identify the core principles of being a responsible digital citizen
- Connect digital citizenship concepts to real-life village community values

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
15-25 min	Explain	Define each category with examples: OS manages hardware, applications do specifi...
25-35 min	Elaborate	Task: 'Create a software guide for a new student. List 5 essential software for ...
35-40 min	Evaluate	Exit ticket: '1. Name one system software and one application software. 2. What ...

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Software licensing involves numbers: version numbers, pricing, user limits

Science: Antivirus software uses pattern recognition to detect threats — similar to how immune systems work

Language: Writing software reviews requires clear, objective language

Digital Citizen Reminder: Only install software from trusted sources — pirated software often contains viruses.

Dormitory Task (Paper-and-Pencil)

Create a table on paper with 3 columns: Software Name, Category System/Application/Utility, What It Does. Fill in 10 software you know. Star the ones you use daily.

Resources Needed:

- Computer if available
- software list
- categorization worksheet
- review template

Differentiation

Approaching: Use a pre-made matching activity to sort software into categories

On Level: Categorize 15 software examples independently and explain each

Advanced: Compare two operating systems Windows vs Linux and write a recommendation

SEND: Card-based sorting activity with teacher guidance, simplified categories

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Pattern Recognition — identifying which category a software belongs to based on its function

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Online Safety Rules

Learning Objectives

- Learn essential online safety rules for internet use
- Identify common online dangers and how to avoid them
- Create a personal online safety checklist

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
15-25 min	Explain	Define each category with examples: OS manages hardware, applications do specifi...
25-35 min	Elaborate	Task: 'Create a software guide for a new student. List 5 essential software for ...
35-40 min	Evaluate	Exit ticket: '1. Name one system software and one application software. 2. What ...

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Password Security

Learning Objectives

- Understand why passwords are important for online security
- Learn to create strong, memorable passwords
- Practice good password management habits

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
15-25 min	Explain	Define each category with examples: OS manages hardware, applications do specifi...
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35-40 min	Evaluate	Exit ticket: '1. Name one system software and one application software. 2. What ...

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Cyberbullying

Learning Objectives

- Define cyberbullying and recognize its various forms
- Understand the emotional and social impact of cyberbullying
- Learn strategies to prevent and respond to cyberbullying

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
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35-40 min	Evaluate	Exit ticket: '1. Name one system software and one application software. 2. What ...

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Digital Footprint

Learning Objectives

- Understand what a digital footprint is and how it's created
- Recognize the permanence of online actions and their consequences
- Learn to manage and protect their digital footprint

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Online Predators

Learning Objectives

- Understand who online predators are and how they target children
- Recognize warning signs of predatory behavior online
- Learn protective strategies to stay safe from online predators

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
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Differentiation

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On Level: Categorize 15 software examples independently and explain each

Advanced: Compare two operating systems Windows vs Linux and write a recommendation

SEND: Card-based sorting activity with teacher guidance, simplified categories

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Pattern Recognition — identifying which category a software belongs to based on its function

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Copyright Basics

Learning Objectives

- Understand what copyright is and why it exists
- Recognize copyright symbols and their meaning
- Learn how to use copyrighted material legally

Timing Breakdown

0-5 min	Engage	Ask: 'What software do you use every day? What would happen if all software disa...
5-15 min	Explore	Show categories of software: system OS, drivers, application browser, editor...
15-25 min	Explain	Define each category with examples: OS manages hardware, applications do specifi...
25-35 min	Elaborate	Task: 'Create a software guide for a new student. List 5 essential software for ...
35-40 min	Evaluate	Exit ticket: '1. Name one system software and one application software. 2. What ...

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.’

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?’

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Software licensing involves numbers: version numbers, pricing, user limits

Science: Antivirus software uses pattern recognition to detect threats — similar to how immune systems work

Language: Writing software reviews requires clear, objective language

Digital Citizen Reminder: Only install software from trusted sources — pirated software often contains viruses.

Dormitory Task (Paper-and-Pencil)

Create a table on paper with 3 columns: Software Name, Category System/Application/Utility, What It Does. Fill in 10 software you know. Star the ones you use daily.

Resources Needed:

- Computer if available
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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Plagiarism

Learning Objectives

- Define plagiarism and understand why it's wrong
- Learn proper ways to credit original sources
- Practice paraphrasing and citing sources correctly

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Phishing Scams

Learning Objectives

- Understand what phishing is and how it works
- Identify common phishing tactics and warning signs
- Learn how to protect themselves from phishing attempts

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Screen Time Management

Learning Objectives

- Understand the effects of excessive screen time on health
- Learn strategies for balanced technology use
- Create a personal screen time management plan

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Safety Poster

Learning Objectives

- Apply digital citizenship knowledge to create an educational poster
- Design visual content that communicates safety messages effectively
- Present and explain their poster to classmates

Timing Breakdown

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 6 Review

Learning Objectives

- Review and consolidate all digital citizenship concepts from Chapter 6
- Demonstrate understanding through a comprehensive review activity
- Create a personal action plan for responsible digital citizenship

Timing Breakdown

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What to improve:

Connection to next lesson: